25901293 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEGEO202 - GROUNDWATER HYDROLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer in 60 words

Answer any **5** questions

Each question carries **2** marks

1. Define vertical distribution of groundwater. [K] / [CO1]
2. State the role of GIS in remote sensing. [K] / [CO2]
3. List two chemical parameters for drinking water. [K] / [CO3]
4. Classify groundwater contaminants. [U] / [CO4]
5. State two limitations of artificial recharge. [K] / [CO4]
6. List the main components of the hydrologic cycle. [K] / [CO1]

[2x5 = 10]

Part B

Short Notes in 250 words

Answer any **6** questions

Each question carries **8** marks

7. Describe SP method in the context of surface geophysical method. [U] / [CO2]

8. Discuss the physical, chemical, and bacteriological standards specified by both WHO guidelines and BIS for potable water quality. [U] / [CO3]

9. Explain the relationship between groundwater overexploitation, hydraulic gradient changes and saline water intrusion in coastal and other aquifers, and describe suitable prevention measures. [U] / [CO4]

10. Discuss the variations in groundwater quality across different Indian provinces. [U] / [CO5]

11. Illustrate how secondary porosity affects water occurrence in hard rocks. [U] / [CO1]

12. Describe briefly the various types of sub-surface geophysical method. [U] / [CO2]

13. Explain how high SAR values cause soil dispersion and reduced permeability. [U] / [CO3]

14. Describe radioactive contamination in groundwater and explain the factors influencing its occurrence and distribution. [U] / [CO4]

[8x6 = 48]

Part C

Essays in 400 words

Answer any **1** questions

Each question carries **12** marks

15. Categorize groundwater basins and explain the hydrological boundaries that define a basin. [U] / [CO1]

16. Describe the process of liquid waste disposal and explain how it degrades the quality of surrounding groundwater. [U] / [CO4]

[12x1 = 12]